



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : OEC- CS801A Big Data Analysis

UPID : 008303

CS/B.TECH(N)/EVEN/SEM-8/8303/2025-2026/1059

Time Allotted : 3 Hours

Full Marks :70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is MRUnit?
- ✓(ii) What company originally developed HBase?
- (iii) Name one popular web analytic tool.
- (iv) Name one industry that heavily uses big data analytics.
- (v) What method is used to open a file in the Hadoop File System API?
- ✓(vi) What is YARN?
- ✓(vii) How does Cassandra achieve high availability?
- (viii) What role does cloud computing play in the convergence of big data trends?
- (ix) How does sharding improve write performance?
- ✓(x) What is replication factor in HDFS?
- (xi) What is the role of the NodeManager In YARN task execution?
- ✓(xii) What is Pig Latin?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- ✓2. What advantages does the document model offer over the key-value model for complex queries? [5]
3. Explain the architecture of HDFS including NameNode and DataNodes. [5]
4. Compare the architecture of Classic MapReduce and YARN. [5]
- ✓5. How does HBase handle row-level atomic operations? [5]
6. How does data locality optimization improve Hadoop job performance? [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- ✓7. (a) Write a detailed note on HBase architecture. [7] ✓
- (b) Describe the roles of HMaster, RegionServer, HDFS, and ZooKeeper. [8] ✓
- ✓8. (a) Discuss in detail why big data has become a strategic priority for businesses. Include real-world benefits, competitive advantages, and potential risks of ignoring big data trends. [9] ✓
- ✓(b) Write an explanatory note on Crowd Sourcing Analytics. [6] ✓
9. (a) Explain the role of a combiner in optimizing MapReduce performance. [5]
- ✓(b) Explain the concept of sharding with a simple example. [5] ✓
- (c) Describe master-slave replication and its advantages. [5]
10. (a) Describe the read and write data flows in HDFS. Include the role of block replication and fault tolerance. [9]
- (b) Explain how Apache Avro integrates with the Hadoop ecosystem. [6]
- ✓11. (a) Explain the architecture of Apache Pig. [6]
- ✓(b) Describe the execution flow of a Pig script. [4] ✓
- ✓(c) Differentiate between HiveQL and traditional SQL [5] ✓

*** END OF PAPER ***